

Second Skin

Project description

Second Skin is a wall-based mechatronic installation made up of three panels, each hiding a 3D-printed body part (a hand, a face, and an ear) underneath a stretched skin of kombucha leather or latex. When a visitor walks into the corner where the panels are grouped, proximity sensors trigger linear actuators that slowly push each form outward through the skin, as if the body parts beneath are trying to break through. When the visitor steps away, the forms sink back under the surface.

The work explores the blurry line between the biological and the man-made. The visitor's living body becomes a kind of fuel. Their closeness is what wakes the synthetic forms up, as if their presence is feeding or calling to them. Each body part means something different: the hand reaches out and grasps, the face stares back and demands to be seen, the ear listens. Together they feel like a body built from parts, waiting just below the surface of something that was once alive.

The skin itself is made from kombucha leather, a material grown through fermentation that sits somewhere between living and manufactured. It is not purely synthetic. It was alive once. What pushes through it is not.

The piece is about what it means to treat the body as a resource, to harvest it, copy it, or keep parts of it alive after the fact. It is in conversation with Stelarc's idea that the human body is becoming outdated, and with Geumhyung Jeong's work around prosthetics as objects that are strange, dependent, and somehow wanting.

Bibliography

Guo, Sophie. "Geumhyung Jeong's Uncanny Anatomy Lesson." ArtReview, 20 January 2025, artreview.com/geumhyung-jeong-under-construction-ica/

Stelarc. "Ear on Arm: Engineering Internet Organ." stelarc.org, 2006 onwards, stelarc.org/_activity-20242.php

Yi, Anicka. Of All Things Orange or Macedonian Wine. 2015. Kombucha leather, metal, paint, stainless steel. Anicka Yi Studio, anickayistudio.biz/artworks/of-all-things-orange-or-macedonian-wine

Materials

- Latex/Kombucha leather pieces x3
- 3D printed body parts

- Face x1
- Ear x1
- Hand x1
- Stepper motors x3
- Wooden boxes x3
- Ultrasonic Sensor x1

Dimensions

Three wall-mounted panels clustered in a corner of the gallery. Each panel is approximately 40 × 40 cm and protrudes ~10 cm from the wall when the actuator is fully extended. The panels are arranged at varied heights.

References/Inspiration

Stelarc - *Ear on Arm*

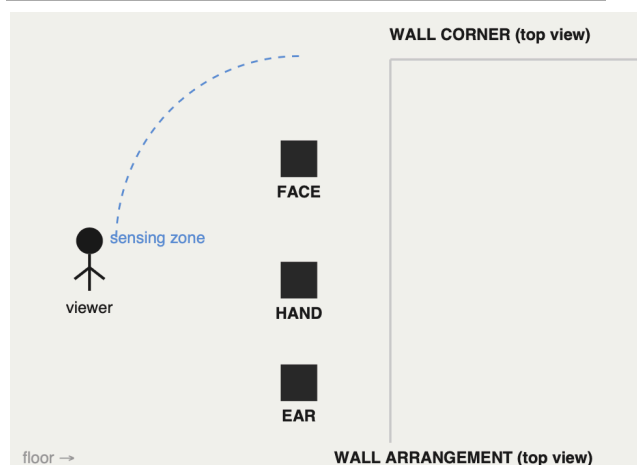
Stelarc surgically implanted a third ear onto his forearm, intended to stream sound to anyone online. The ear exists for others, not for him. This idea of body parts removed from their original context and repurposed for a different function is a direct influence on Second Skin.

Geumhyung Jeong - *Uncanny Anatomy Lesson*

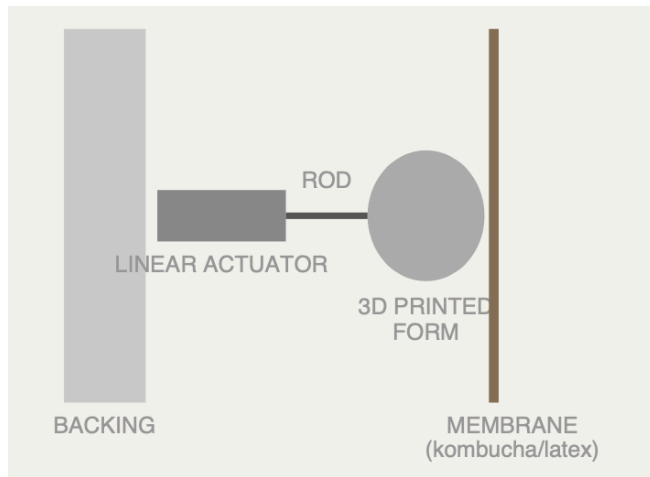
Geumhyung Jeong builds and performs with handmade mechanical puppets and prosthetics that mimic human movement in deeply uncomfortable ways. Her work treats the body as something that can be assembled, rehearsed, and replaced. This tension between the constructed and the living directly shapes how Second Skin thinks about synthetic body parts wanting to be real.

Sketches

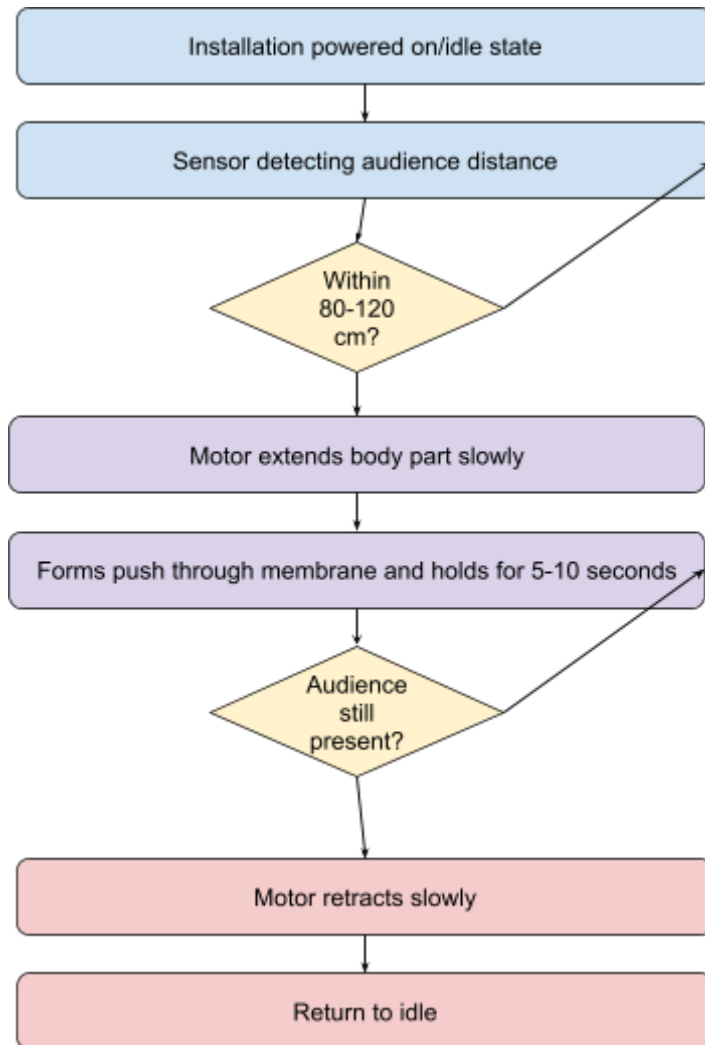
Overview of what installation would look like:



Overview of mechanism behind each panel:



Interaction Flow Chart



Budget

Item	Cost	Source
Kombucha leather	\$30-60	Amazon
Latex	\$25	Amazon/Etsy
Stepper motor	0	Provided by DXARTS class, Amazon if needed
Wooden boxes	\$20-50	Home depot for wood (if needed)
Ultrasonic sensor	0	Provided by DXARTS class, Amazon if needed
3D Printed body parts	0	3D printers and material at The8
Total: \$75 - 135		